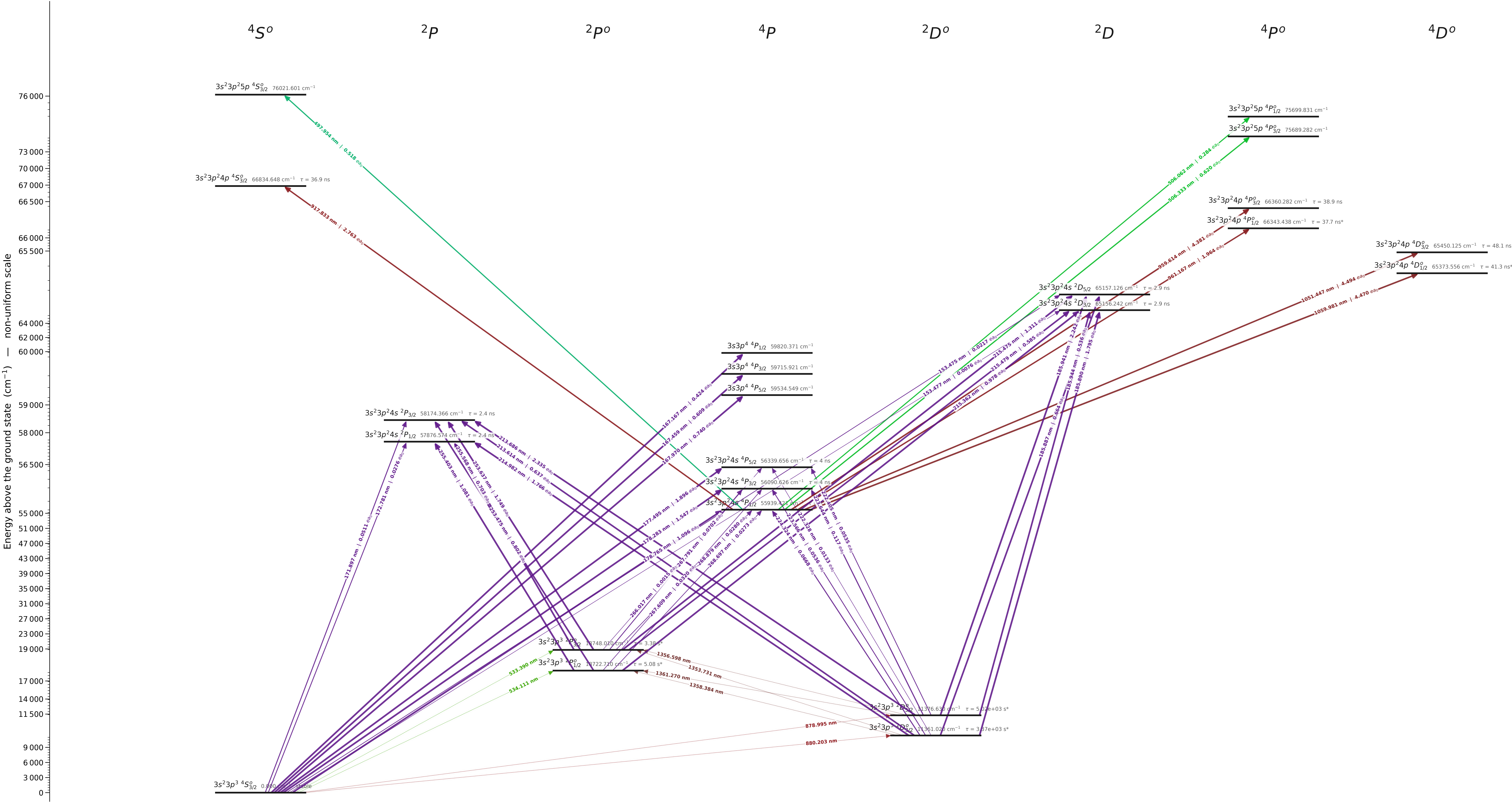
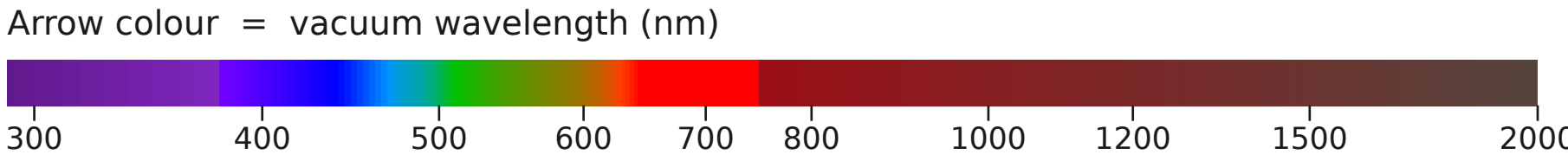


Phosphorus

P (P I). Strongest classified transitions below 2 μm (NIST): 23 levels, 50 lines.

P⁺ ionisation limit 84580.83 cm⁻¹ = 10.48669 eV (118.230 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $|U||er||J' \rangle|$ in ea₀
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	⟨J er J'⟩ (ea ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
3s²3p³ 4S° _{3/2} - 3s3p⁴ 4P _{1/2}	167.1671	-	1793.3696	0.424	3.900e+07	-	-	167.2 / 167.5 / 168.0 nm multiplet 3p3 4S°3/2 - 3s3p4 4P
3s²3p³ 4S° _{3/2} - 3s3p⁴ 4P _{3/2}	167.4595	-	1790.2383	0.609	4.000e+07	-	-	167.2 / 167.5 / 168.0 nm multiplet 3p3 4S°3/2 - 3s3p4 4P
3s²3p³ 4S° _{3/2} - 3s3p⁴ 4P _{5/2}	167.9697	-	1784.8009	0.740	3.900e+07	-	-	167.2 / 167.5 / 168.0 nm multiplet 3p3 4S°3/2 - 3s3p4 4P
3s²3p³ 4S° _{3/2} - 3s²3p²4s 4P _{3/2}	177.4949	-	1689.0204	1.896	2.170e+08	39.79	86.8	177.5 / 178.3 / 178.8 nm resonance triplet 3p3 4S°3/2 - 4...
3s²3p³ 4S° _{3/2} - 3s²3p²4s 4P _{3/2}	178.2829	-	1681.5547	1.547	2.140e+08	39.79	85.6	177.5 / 178.3 / 178.8 nm resonance triplet 3p3 4S°3/2 - 4...
3s²3p³ 4S° _{3/2} - 3s²3p²4s 4P _{1/2}	178.7648	-	1677.0216	1.096	2.130e+08	39.79	85.2	177.5 / 178.3 / 178.8 nm resonance triplet 3p3 4S°3/2 - 4...

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).

³¹P hyperfine structure (I = 1/2)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
3s²3p³ 4S° _{3/2}	55.055691(8)	-	F=1: -68.82 F=2: +41.29

Measured A, B constants; sources in the data file.